Minimize:

$$+1\alpha_1 + 2\alpha_1\alpha_2 - 5.5\alpha_1\alpha_3 - 6\alpha_1\alpha_4 + 2\alpha_2\alpha_3 + 4\alpha_2 - 11\alpha_2\alpha_4 - 12\alpha_3\alpha_4$$

$$-5.5\alpha_1\alpha_3 - 11\alpha_1\alpha_4 + 32.5\alpha_2^2 + 36\alpha_3\alpha_2 - 6\alpha_3\alpha_4 - 12\alpha_3\alpha_2 + 36\alpha_3\alpha_4 + 40\alpha_4^2] - [\, \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 \,]$$

Subject to:

$\sum_i \alpha_i y_i = 0$

$\alpha_1 + \alpha_2 - \alpha_3 - \alpha_4 = 0$

$\alpha_i \geq 0$

$\alpha_1 \geq 1,\ \alpha_2 \geq 1,\ \alpha_3 \geq 1,\ \alpha_4 \geq 1$

$\alpha_i \leq C$

$\alpha_1 \leq C,\ \alpha_2 \leq C,\ \alpha_3 \leq C,\ \alpha_4 \leq C$

Quadratic Equation in α with constraints

$\Rightarrow$

$-\alpha_1 \geq 0,\quad -\alpha_2 \geq 0,\quad -\alpha_3 \geq 0,\quad -\alpha_4 \geq 0$

$\alpha_1 \geq 1,\ \alpha_2 \geq 1,\ \alpha_3 \geq 1,\ \alpha_4 \geq 1$

$G\ x \geq h$ $\quad$ $A\ x = b$

G =

$$\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{pmatrix}$$

h =

$$\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$A = [-1 \quad -1 \quad -1 \quad -1]_{1\times 4} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{pmatrix}$$

$$b = [0]_{1\times 1}$$

Solution

$\alpha_1 = 0.0000000002,\ \alpha_2 = 0.0000000001,$
$\alpha_3 = 0.0000000001,\ \alpha_4 = 0.0000000000$